

WTO Accession, PNTR, and U.S. Import Quantities and Prices

w26-causal-inference-economics project deck (12 minutes)

Vincent Zheng

University of Chicago

February 27, 2026

Talk Roadmap (12 min)

- Research question and policy context (2 min)
- Data and empirical strategy (4 min)
- Main results and event-study evidence (4 min)
- Limitations, robustness, and next steps (2 min)

Research Question and Contribution

Core question

How did the resolution of U.S. trade-policy uncertainty around China (PNTR in 2000, WTO accession in 2001/2002 implementation) affect:

- import quantities into the U.S., and
- import prices paid by U.S. buyers?

Contribution in this project

- Reproducible monthly panel pipeline at product level.
- Difference-in-differences and event-study estimates tied to NTR-gap exposure.
- Separate analysis for value, quantity, and unit-value outcomes.

Policy Timing and Identification Intuition

- Pre-2001: China receives renewable NTR treatment; non-renewal risk implies tariff-jump uncertainty.
- 2000: U.S. grants PNTR (legal permanence conditional on WTO accession).
- Dec 2001 / Jan 2002: WTO accession and operational transition.

Identification idea

Products with larger pre-period NTR gaps should respond more after policy uncertainty is resolved.

$$\text{Exposure}_p = \text{gap_pre}_p = (\text{non-NTR tariff})_p - (\text{NTR tariff})_p$$

Data and Sample

Data sources

- USITC monthly imports by product and partner (pipeline in `data_cleaning_scripts/`).
- Tariff schedule information used to construct product-level `gap_pre`.

Panel and outcomes

- Product-month panel with high-dimensional fixed effects.
- Outcomes: `ln_customs_value`, `ln_first_unit_qty`, `ln_unit_value`.
- Main output files generated under `analysis/results/`.

Sample scale from current runs

- DiD common sample: about 451k product-month observations.
- Continuous-treatment sample: up to about 767k observations.

Baseline Specification

Two-way FE DiD

$$Y_{pt} = \beta (\text{HighGap}_p \times \text{Post}_t) + \delta_p + \lambda_t + \varepsilon_{pt}$$

- Y_{pt} : log import value, log quantity, or log unit value.
- δ_p : product fixed effects.
- λ_t : time fixed effects.
- SEs clustered at product level.

Continuous variant

$$Y_{pt} = \theta (\text{gap_pre}_p \times \text{Post}_t) + \delta_p + \lambda_t + \varepsilon_{pt}$$

Event-Study Design

- Replace one post indicator with event-time leads/lags.
- Reference period: month before policy window (`event_time = -1`).
- Estimation window in code: 60 months pre and 48 months post.

Use in the talk

- Check pre-trend behavior.
- Show timing of quantity/value response.
- Compare price dynamics (`ln_unit_value`) against quantity dynamics.

Main DiD Results (Current Outputs)

Binary treatment (HighGap x Post) with product and date FE

- $\ln_customs_value$: $\hat{\beta} \approx 0.121$ (s.e. 0.031), positive and significant.
- $\ln_first_unit_qty$: $\hat{\beta} \approx 0.145$ (s.e. 0.036), positive and significant.
- $\ln_unit_value$: $\hat{\beta} \approx -0.024$ (s.e. 0.017), negative but not strongly significant in FE spec.

Interpretation

- High-gap products show stronger post-policy growth in value and quantity.
- Price effects are weaker in the binary split and clearer in continuous treatment.

Continuous-Treatment Results

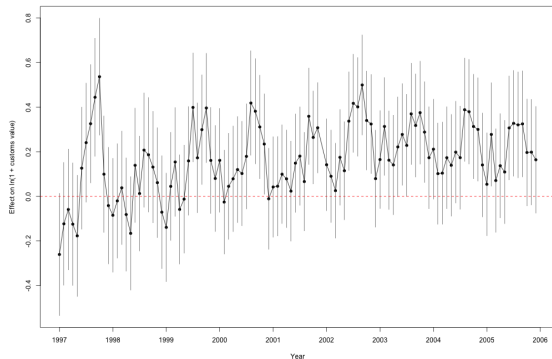
From `did_continuous_gap_pre_post_effects.csv`:

- Value (`ln_customs_value`): $\hat{\theta} = 0.122$ (s.e. 0.134).
- Quantity (`ln_first_unit_qty`): $\hat{\theta} = -0.061$ (s.e. 0.130).
- Unit value (`ln_unit_value`): $\hat{\theta} = -0.131$ (s.e. 0.032), strong negative effect.

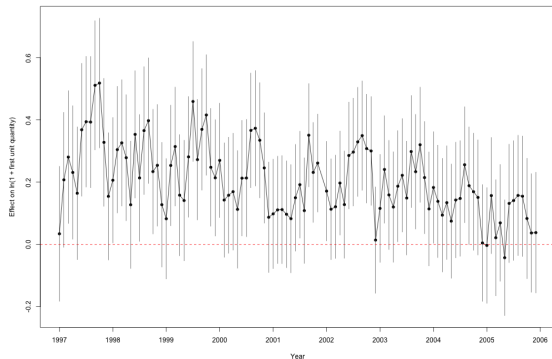
Reading

Higher tariff-jump exposure is associated with stronger post-policy price declines, consistent with lower import prices in more exposed products.

Event Study: Value and Quantity

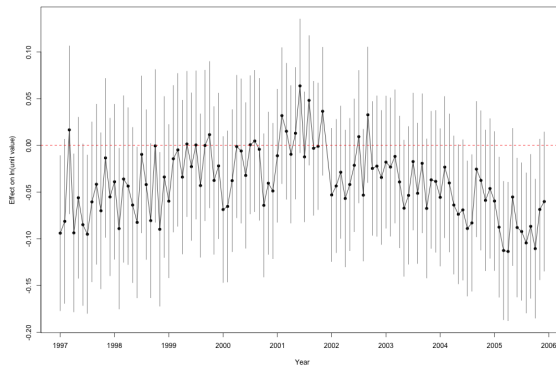


Value event study



Quantity event study

Event Study: Unit Value (Price Proxy)



Continuous-treatment pre-trend test for unit value: $p \approx 0.081$ (marginal).

Assumptions, Risks, and Robustness Plan

Key assumptions

- Parallel trends between more/less exposed products.
- No confounding shocks perfectly aligned with exposure and timing.

Current risks

- Strong pre-trend rejections in some continuous event-study outcomes.
- Potential measurement noise in unit values and quantity units.

Planned robustness checks

- Alternative post definitions and narrower windows.
- Add controls for seasonality and unit-consistency filters.
- Heterogeneity by Rauch classification and sector exclusions.

Takeaways and Next Steps

Takeaways

- Evidence supports a post-policy increase in import value and quantity for high-gap products.
- Price effects are more visible under continuous exposure than binary treatment.

Immediate next steps before final presentation

- Tighten identification and pre-trend diagnostics.
- Convert core regression outputs into one polished summary table.
- Add one motivation figure and one conclusion figure for a cleaner narrative.

Questions